	Part grounding (target task)							Object and referring grounding (reference)					Rejection (reference)	
	PARTONOMY		GSEval	PACO	PPS	PartImage-	Instruct-	PACO	PPS	GSEval	gRefCOCO		gRefCOCO N-acc	
Method	micro	macro	Part	Part[§]	-116[§]	Net	Part	Object[§]	Object[§]	All	testA[§]	testB[§]	testA[§]	testB[§]
Segmentation pipelines (no MLLM)														
SAM 3 (text)	30.5	31.7	30.7	25.8	30.0	24.3	66.5	75.1	77.5	42.4	42.2	47.8	89.1	82.9
Grounded-SAM 2	25.9	29.0	28.0	27.2	17.7	37.2	40.4	64.2	79.9	62.9	48.0	42.3	21.4	19.1
MLLM backbone: Qwen3-VL-4B														
Sa2VA	41.0	43.0	21.1	19.7	19.9	40.1	56.5	46.8	78.5	63.4	51.5	46.0	19.0	17.6
Box + SAM 3	47.1	48.8	37.5	33.0	33.6	45.1	67.5	76.4	80.0	**72.2**	50.7	48.0	27.7	32.4
PACT (Ours)	**49.4**	**51.0**	**51.4**	**47.5**	**40.6**	**50.7**	**68.2**	**81.8**	**80.6**	70.0	**59.5**	**55.9**	**94.0**	**90.9**
Δ vs. best row above	+2.3	+2.2	+13.9	+14.5	+7.1	+5.6	+0.8	+5.4	+0.7	−2.3	+8.0	+7.9	+4.9	+8.1

Table 1: **Controlled comparison** on test splits (gIoU, %; N-acc: accuracy on no-target expressions). Every row is run by us with the same queries and scorer, Qwen3-VL-4B wherever an MLLM is used, and SAM 3 as mask source (Sa2VA: its own decoder; Grounded-SAM 2: Grounding DINO boxes + SAM 2). Every method outputs at most one mask per query: its top-scoring instance, or none below its threshold (PACT on gRefCOCO: threshold chosen on val). With all instances merged, SAM 3 reaches 53.9 on PartImageNet (as reported by OP-HRG). The pipelines run at their released thresholds; without its score threshold SAM 3 reaches 41.9 on PACO Part and 82.1 on PACO Object. Δ: per column, PACT minus the best row above (from unrounded values); bold: best. Every Δ differs from zero under a paired image-level bootstrap ($p < 10^{-4}$) except InstructPart ($p = 0.36$) and PPS-116 Object ($p = 0.03$). Results with 8B and 32B backbones are in the appendix (Tab. 7). [§]our model trains on this dataset's training split; all other columns are zero-shot for every row.

		Parameters				Part grounding					Object and referring			Rejection	
						PARTONOMY	GSEval	PPS	PartImage-	Instruct-	GSEval	gRefCOCO		gRefCOCO N-acc	
Method	Size	Total	Trainable	%	Training data	macro	Part	-116	Net	Part	All	testA	testB	testA	testB
(A) A `[SEG]` token decodes the mask															
UniPixel	3B	3.88B	-	-	∼2.9M samples	–	–	30.6	46.2	59.7	–	–	–	–	–
InstructSeg	3B	-	-	-	RefCOCO/+/g, ReasonSeg, video	–	24.2	–	–	–	52.5	–	–	–	–
Sa2VA	4B	3.94B	0.88B[‡]	22%	1.1M samples	–	17.4	14.7	38.1	50.2	58.5	–	–	–	–
Sa2VA	8B	8.32B	0.37B[‡]	4.4%	1.1M samples	–	18.0	–	–	–	56.3	–	–	–	–
LISA	7B	-	-	-	10 datasets	–	21.2	13.8[§]	29.9[§]	43.3	57.6	–	–	–	–
LISA (GRES-ft)	7B	-	-	-	10 datasets + gRefCOCO	–	–	–	–	–	–	66.3[§]	58.8[§]	50.0[§]	51.9[§]
GLaMM	7B	8.02B	0.59B[‡]	7.3%	GranD (11M images)	–	16.5	–	–	–	57.2	–	–	–	–
SegLLM	7B	-	-	-	MRSeg	32.4	–	–	–	–	–	–	–	–	–
PLUM (PARTONOMY-ft)	13B	-	-	-	mixed + PARTONOMY	41.6[§]	–	–	–	–	–	–	–	–	–
(B) Masks as discrete tokens															
SAMTok	3B	-	-	-	5M SFT + RL	–	40.8	–	–	–	67.8	–	–	–	–
SAMTok	4B	5.06B	0.91B[‡]	18%	5M SFT + RL	–	–	–	–	–	–	76.9[§]	69.1[§]	66.2[§]	60.8[§]
SAMTok (GRES-ft)	7B	-	-	-	5M SFT + RL + gRefCOCO	–	–	–	–	–	–	78.5[§]	69.6[§]	75.9[§]	68.6[§]
(C) The MLLM prompts a frozen segmenter															
OP-HRG	4B	4.66B	4.44B	95%	8.2K pairs (RL)	–	–	38.6	**56.9**	**75.6**[§]	–	–	–	–	–
VisionReasoner	7B	8.52B	-	-	7K samples (RL)	–	–	27.4	45.6	59.4	–	–	–	–	–
(D) Selection over mask proposals															
PANORAMA	4B	5.32B	1.32B	25%	957K samples	–	**53.5**	–	–	–	**75.5**	**79.8**[§]	**74.5**[§]	85.5[§]	81.2[§]
PACT (Ours)	4B	5.36B	**64.2M**	**1.2%**	PACO, Pascal-Part, gRefCOCO	**51.0**	51.4	**40.6**[§]	50.7	68.2	70.0	72.3[§]	65.2[§]	**89.6**[§]	**86.0**[§]

Table 2: **Comparison with published systems**, grouped by how the mask is produced (test splits; gIoU, %; N-acc as in Tab. 1). Numbers as published under the same protocol; our gRefCOCO cells use that protocol's multi-mask readout (threshold from val), Tab. 1 the one-mask readout. Size: backbone size; rows larger than 4B are references, and bold marks the best result with ≤4B. Parameters: total at inference including any frozen segmenter (model cards), trainable as printed in the paper, and their ratio (ours: Qwen3-VL-4B + SAM 3 + 64.2M trained, counted exactly from the model code). [‡]measured by PANORAMA under each method's released recipe (its Tab. 17). PPS-116: Pascal-Part-116. "-" / "–": not reported. [§]the method trains on this benchmark's training split (in-domain); all other cells are zero-shot.

		Part grounding						Object and referring			Rejection	
		PARTONOMY	GSEval	PACO	PPS	PartImage-	Instruct-	GSEval	gRefCOCO		gRefCOCO N-acc	
	Candidates → selection	macro	Part	Part§	-116§	Net	Part	All	testA§	testB§	testA§	testB§
(a)	SAM 3 text only (its own ranking)	31.7	30.7	25.8	30.0	24.3	66.5	42.4	42.2	47.8	89.1	82.9
(b)	grid pool (query-free) → mask tokens	44.8	48.5	35.9	29.4	43.9	61.1	59.2	57.5	54.8	90.4	88.0
(c)	query pool → pooled features (no mask tokens)	44.8	33.8	42.4	35.2	48.2	64.0	45.4	58.7	55.1	85.9	84.3
(d)	query pool → mask tokens, free attention	50.4	**57.8**	47.1	**40.7**	50.4	**68.3**	**71.9**	55.7	55.2	93.4	**91.2**
(e)	PACT: query pool → region-restricted mask tokens	**51.0**	51.4	**47.5**	40.6	**50.7**	68.2	70.0	**59.5**	**55.9**	**94.0**	90.9

Table 3: **Ablations** on test splits (gIoU, %; N-acc: accuracy on no-target expressions). (a)→(b): the MLLM selecting among candidates instead of SAM 3's own ranking; (b)→(e): query-conditioned candidates; (c)→(e): candidates as mask tokens inside the MLLM instead of pooled features; (d) vs (e): free attention helps GSEval, while restricting each mask token to its region gives better grounding on gRefCOCO (the variant chosen on validation). Every row outputs at most one mask per query. Rows (b)–(e) use the same data, schedule and backbone; checkpoint, layer and part-decision rule of each are chosen on validation. §in-domain for (b)–(e).

	gIoU			Gap	Pick vs. SAM 3 top-1	
	SAM 3 top-1	PACT pick	Best in pool	closed	PACT better	SAM 3 better
PARTONOMY	42.0	**49.4**	71.0	26%	**25.9%**	11.5%
InstructPart	67.4	**68.2**	80.0	7%	11.3%	**11.8%**
	Source of PACT's pick				Source of the best mask	
	point grid	SAM 3 part	SAM 3 object		point grid	SAM 3 text
PARTONOMY	35.3%	57.7%	7.0%		58.8%	41.2%
InstructPart	28.0%	65.2%	6.8%		52.3%	47.7%

Table 4: **Selection versus coverage** on part queries (test splits). *Best in pool*: an oracle that picks the candidate with the highest IoU. *Gap closed*: share of the distance from SAM 3's own top instance to the oracle that PACT's selection recovers. *Better*: queries where one pick beats the other by at least 0.1 IoU. The candidate pool contains a much better mask than either pick (71.0 vs. 49.4 on PARTONOMY), so the remaining errors are selection errors; most of the best masks come from the query-free point grid, which SAM 3's text instances alone would miss.

Method	InstructPart (part-level)				ReasonSeg (object-level)	
	human instr.		GPT-4 instr.		test	
	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU
Segmentation pipeline (no MLLM)						
SAM 3 (text), released threshold	1.6	4.2	11.4	10.7	8.3	11.6
SAM 3 (text), always answers	30.6	20.8	26.2	19.0	31.3	27.2
MLLM backbone: Qwen3-VL-4B						
Sa2VA	41.3	36.7	44.7	36.9	54.5†	**50.3**†
Box + SAM 3	49.5	41.4	52.4	42.4	**60.4**	49.5
PACT (Ours)	50.0	37.7	53.3	40.9	50.7	41.3
PACT + query rewriting (Ours)	**60.8**	**51.4**	**61.0**	**53.0**	59.5	45.9
Δ vs. Box + SAM 3	+11.2	+10.0	+8.5	+10.6	−0.9	−3.6

Table 5: **Reasoning queries** (test splits; %). Every method outputs at most one mask per query; nothing is tuned on these benchmarks. Baselines use their own instruction or question prompt (phrase prompts in the appendix); PACT reads the query as given. *Query rewriting*: the pool's SAM 3 phrase is the target named by the base Qwen3-VL-4B (the backbone of PACT and of Box + SAM 3), a rule fixed on ReasonSeg val before any test run. Paired image-level bootstrap against Box + SAM 3: InstructPart gIoU +11.2 / +8.5 ($p < 10^{-4}$), cIoU +10.0 / +10.6 ($p = 0.002$); ReasonSeg gIoU −0.9 ($p = 0.39$), cIoU −3.6 ($p = 0.31$). Δ: PACT + query rewriting minus Box + SAM 3, the same-backbone pipeline; bold: best. †Sa2VA trains on ReasonSeg's training split; all other cells are zero-shot.

	GEdit-Bench-EN						ImgEdit-Bench					
	Mask		Qwen-Image-Edit		FLUX Fill		Mask		Qwen-Image-Edit		FLUX Fill	
Mask source	IoU↑	empty↓	judge↑	LPIPS-out↓	judge↑	LPIPS-out↓	IoU↑	empty↓	judge↑	LPIPS-out↓	judge↑	LPIPS-out↓
free edit (no mask)	–	–	7.31	0.130	–	–	–	–	3.72	0.162	–	–
GT edit region (oracle)	–	–	6.21	0.039	3.16	0.028	–	–	3.49	0.026	1.98	0.017
GT bbox (oracle)	–	–	6.71	0.074	3.96	0.127	–	–	3.59	0.084	1.78	0.153
SAM 3 (text)	52.9	**0.0**	6.17	0.049	3.45	0.074	66.3	**0.0**	**3.40**	**0.031**	2.08	0.027
Box + SAM 3	**57.6**	1.3	6.27	**0.046**	3.31	**0.047**	66.1	2.8	3.34	0.033	2.02	0.029
PACT (Ours)	53.6	**0.0**	**6.45**	0.053	**3.62**	0.091	**67.5**	**0.0**	3.39	0.033	**2.11**	**0.026**

Table 6: **Region-grounded editing** on all GEdit-Bench-EN and ImgEdit-Bench samples (CoCoEdit masked protocol). Each mask source localizes the edit's referent; the editor then edits within that mask (Qwen-Image-Edit: its edit blended into the mask; FLUX Fill: inpainting the mask). Mask IoU and empty rate (%) against the GT edit region; judge: Qwen2.5-VL-72B on the benchmark's scale; LPIPS-out: change outside the GT edit region. The judge favours larger edit regions (the GT bbox scores above the exact GT region), so judge differences between mask sources are not read as edit quality. PACT returns a mask for every request. Bold: best of the three mask sources.

	Part grounding							Object and referring grounding					Rejection	
	PARTONOMY		GSEval	PACO	PPS	PartImage-	Instruct-	PACO	PPS	GSEval	gRefCOCO		gRefCOCO N-acc	
Method	micro	macro	Part	Part[§]	-116[§]	Net	Part	Object[§]	Object[§]	All	testA[§]	testB[§]	testA[§]	testB[§]
MLLM backbone: Qwen3-VL-4B (main paper, Tab. 1)														
Box + SAM 3	47.1	48.8	37.5	33.0	33.6	45.1	67.5	76.4	80.0	72.2	50.7	48.0	27.7	32.4
PACT (Ours)	49.4	51.0	51.4	47.5	40.6	50.7	68.2	81.8	80.6	70.0	59.5	55.9	94.0	90.9
MLLM backbone: Qwen3-VL-8B														
Box + SAM 3	46.5	48.2	26.3	33.2	32.8	47.0	69.0	75.8	80.1	69.7	50.2	47.4	24.4	29.3
PACT (Ours)[†]	50.0	51.3	50.8	47.4	39.6	50.7	68.8	81.3	80.3	68.8	58.1	56.2	89.2	88.0
Δ vs. Box + SAM 3	+3.5	+3.1	+24.5	+14.3	+6.8	+3.6	−0.1	+5.5	+0.3	−1.0	+7.8	+8.7	+64.8	+58.7
MLLM backbone: Qwen3-VL-32B														
Box + SAM 3	49.3	50.5	–	–	–	–	–	–	–	–	55.3	52.1	44.7	43.8

Table 7: **Larger MLLM backbones** under the protocol of Tab. 1 (test splits; gIoU, %; N-acc: accuracy on no-target expressions). Δ: PACT minus Box + SAM 3 with the same 8B backbone. With 8B, PACT leads Box + SAM 3 on every column except InstructPart (−0.1) and GSEval All (−1.0); compared with the 4B model it is within 1.5 points on every grounding column and lower on rejection (N-acc −4.8 / −2.9). Every method outputs at most one mask per query. The 32B pipeline was run on PARTONOMY and gRefCOCO only. [†]same pool and data as the 4B model, without the tree-relation bias and family loss. [§]PACT trains on this dataset's training split; all other columns are zero-shot.

	InstructPart (human)		InstructPart (GPT-4)		ReasonSeg	
Baseline, prompt	gIoU	cIoU	gIoU	cIoU	gIoU	cIoU
Sa2VA, phrase prompt	43.3	38.8	44.3	36.5	53.9	48.9
Sa2VA, own question prompt	41.3	36.7	44.7	36.9	54.5	50.3
Box + SAM 3, phrase prompt	48.7	41.6	52.4	47.0	58.6	47.6
Box + SAM 3, sentence prompt	49.5	41.4	52.4	42.4	60.4	49.5

Table 8: **Baseline prompts on reasoning queries** (test; %). Phrase prompt: the template used for noun phrases in Tab. 1; own prompt: Sa2VA's question prompt from its evaluation code, Qwen3-VL asked for the region the request refers to (Tab. 5). The comparison with PACT holds under either prompt.

	testA§			testB§		
Method (multi-mask output)	gIoU	cIoU	N-acc	gIoU	cIoU	N-acc
SAM 3 (text), all instances	44.3	29.8	89.3	50.5	39.7	82.8
SAM 3 (text), all instances, threshold from val	47.6	37.1	83.2	51.7	43.8	76.1
Grounded-SAM 2, all boxes	51.5	47.6	28.3	44.7	39.4	28.4
Sa2VA	51.5	51.0	19.0	46.0	46.1	17.6
Box + SAM 3, all boxes	62.9	65.8	27.7	57.1	**58.9**	32.4
PACT (Ours), set of candidates above t	**72.3**	**67.2**	**89.6**	**65.2**	58.3	**86.0**

Table 9: **Multi-target referring** on gRefCOCO (test; %), where an expression may name several objects or none and every method may output several masks (their union is scored). PACT keeps every candidate scoring above a threshold chosen on val ($t = 0.9$); baselines output all instances or boxes above their released thresholds. §PACT trains on gRefCOCO's training split.

	Part grounding						Object and referring			Rejection	
	PARTONOMY	GSEval	PACO	PPS	PartImage-	Instruct-	GSEval	gRefCOCO		gRefCOCO N-acc	
	macro	Part	Part§	-116§	Net	Part	All	testA§	testB§	testA§	testB§
(e′) without tree-relation bias and family loss	50.8	49.1	47.5	39.9	50.3	67.2	**70.1**	**59.6**	**57.1**	89.9	85.9
(e) PACT	**51.0**	**51.4**	47.5	**40.6**	**50.7**	**68.2**	70.0	59.5	55.9	**94.0**	**90.9**

Table 10: **Tree-relation terms** (test; gIoU, %; N-acc: accuracy on no-target expressions). (e′) is (e) of Tab. 3 without the learned attention bias by tree relation between mask tokens and without the family loss; everything else is identical. Grounding stays within about one point; rejection of no-target expressions improves by 4.1 / 5.0 N-acc with the two terms. §in-domain for both rows.